

Supply Chain Simulations Benchmark Report

This report presents a comparative quantitative analysis of the results obtained from four different simulations of a supply chain. Each simulation has generated its results in Excel format, and the key performance indicators (KPIs) have been processed from the 'Table 1' sheet. The table below summarizes the values obtained by KPI and scenario, excluding those with null values in all simulations.

KPIs Comparison Table by Scenario

KPI	1-FLAT DEMAND	Unit	2-NORMAL DEMAND	3-LOADING GATE NORMAL(1,3)	3-LOADING GATE NORMAL(2,6)
Average Cost per Order	341,790.01	USD	346,992.88	350,602.20	355,028.06
Average Cost per Item	850.86	USD	855.95	866.11	871.74
Profit	286,646,904.78	USD	276,172,249.91	251,603,299.13	235,266,104.11
ELT Service Level by Revenue	1.00	Ratio	1.00	0.54	0.50
Facility Cost	82,692,201.60	USD	92,732,050.14	110,957,834.46	123,974,504.54
Inventory Spend	1,444,377,600.00	USD	1,445,517,177.35	1,445,378,284.13	1,453,090,495.99
Bullwhip Effect by Product	0.51	Ratio	2.48	2.12	2.07
Revenue	1,932,024,000.00	USD	1,932,716,241.75	1,926,079,406.89	1,930,880,126.87
Revenue by Customer	1,926,211,200.00	USD	1,927,546,100.83	1,919,842,550.51	1,924,460,099.26
Total Cost	1,645,377,095.22	USD	1,656,543,991.84	1,674,476,107.75	1,695,614,022.76
Transportation Cost	118,307,293.62	USD	118,294,764.35	118,139,989.16	118,549,022.23

Quantitative Analysis by Scenario

Stage: 1-FLAT DEMAND

- Lower costs per order and per unit.

- - Maximum profitability of the system.
- - Perfect customer service level (100%).
- - Minimal installation costs.
- - It does not consider real variations in the demand environment.
- - Poor robustness to unexpected changes or variability.

Scenario: 2-NORMAL DEMAND

- - Perfect level of service despite variability.
- - Contained costs, although higher than in flat demand.
- - Slightly lower profitability.
- - Slight increase in infrastructure costs.

Escenario: 3-LOADING GATE NORMAL(1,3)

- - It allows simulating the impact of the bottleneck on the loading gate.
- - Service level drops to 54%.
- - Significant decrease in profits.
- - High cost of infrastructure.

Escenario: 3-LOADING GATE NORMAL(2,6)

- - Simulation of a demanding logistics environment.
- - Maximum operating and infrastructure costs.
- - Service level below 50%.
- - Minimum profitability between the four scenarios.

Detailed Analysis of Causes of Variation Between Scenarios

The observed variations in the key performance indicators (KPIs) between the different simulations can be attributed to the following key factors:

1. ****Variability in demand****: - In the "1-FLAT DEMAND" scenario, constant demand allows for efficient inventory, production, and transportation planning. - In "2-NORMAL DEMAND", the introduction of fluctuations increases the complexity of the operation, generating slight increases in cost and a slight drop in profitability.
2. ****Loading Gate Restrictions****: - The "3-LOADING GATE NORMAL(1,3)" and "3-LOADING GATE NORMAL(2,6)" scenarios incorporate load point capacity or availability constraints. - This generates bottlenecks that directly impact on: - The level of service (late or incomplete deliveries). - Increased logistics costs due to downtime, rework, or the need for additional resources. - The need to expand facilities or hire more staff to manage congestion.
3. ****Infrastructure Cost****: - As operational complexity increases, Facility Costs increase due to: - Increased use of storage space. - More intensive use of logistical resources. - Need

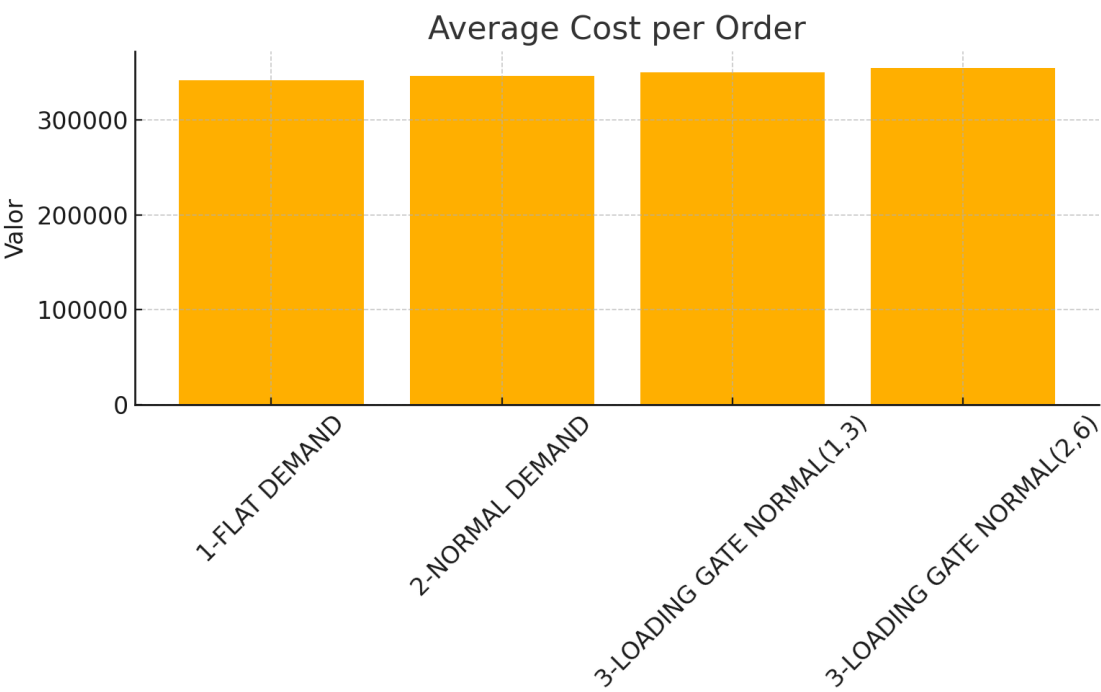
for additional technology or equipment to mitigate delays.

4. ****Impact on profitability****: - The combination of higher logistics costs and lower customer satisfaction (measured by the level of service) produces a drastic drop in profits in the most demanding scenarios. - This underscores the importance of balancing operational efficiency with system resilience.

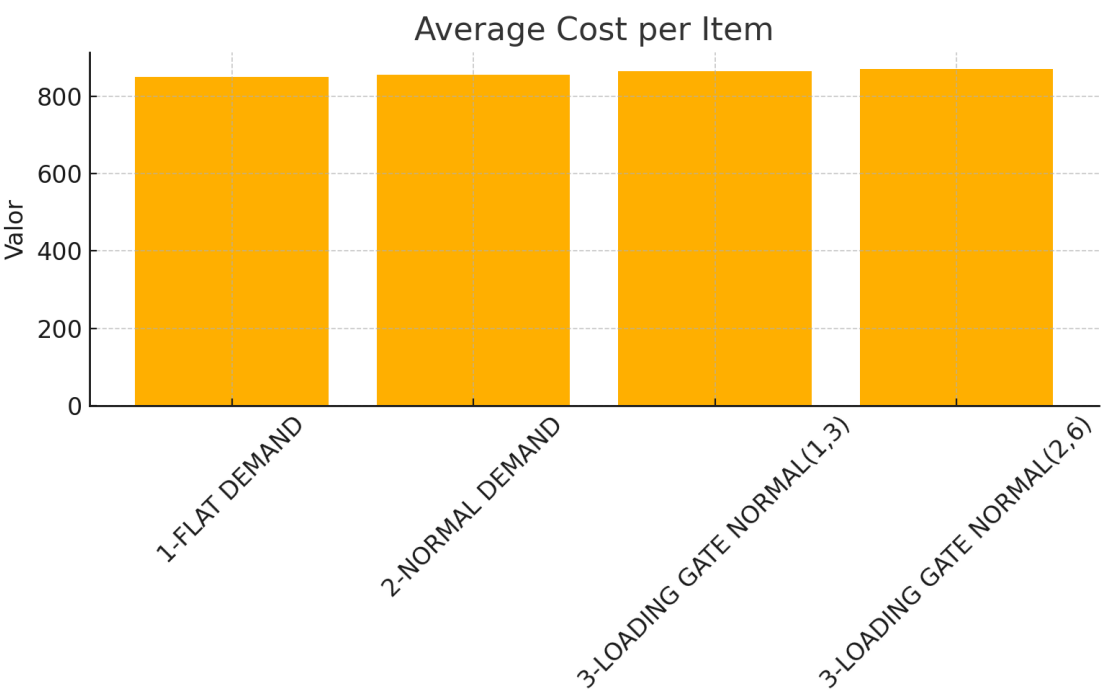
In summary, the results suggest that while a controlled and predictable environment maximizes benefits, it is essential to model real constraints to design more robust and adaptable supply chains.

KPI Comparison Visualization

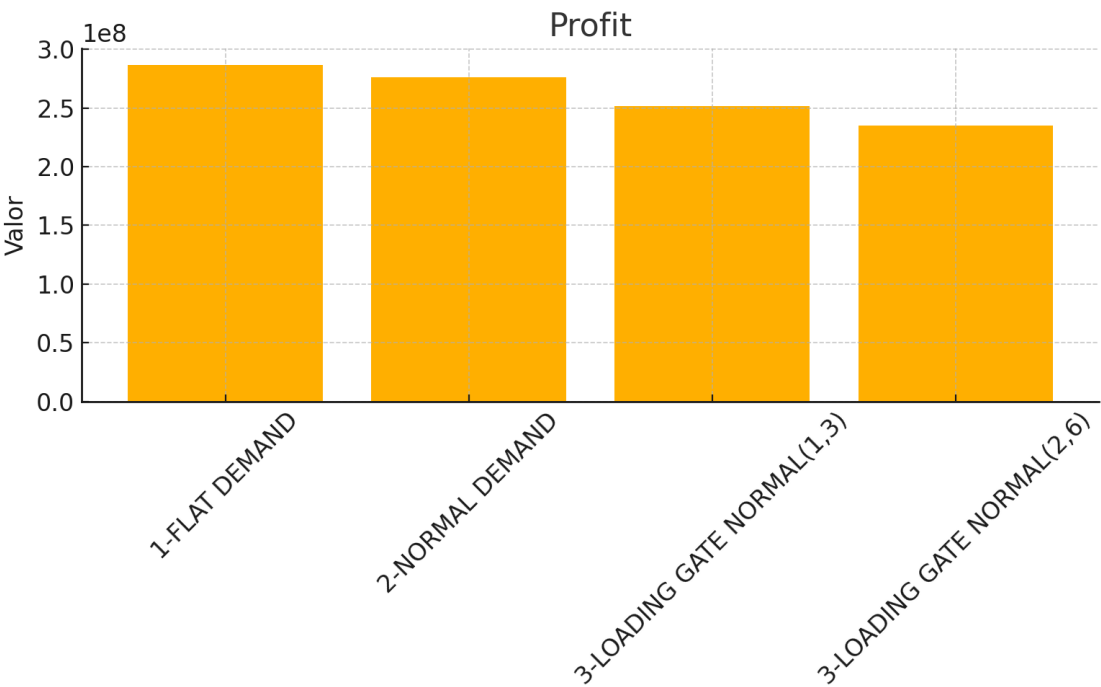
Average Cost per Order



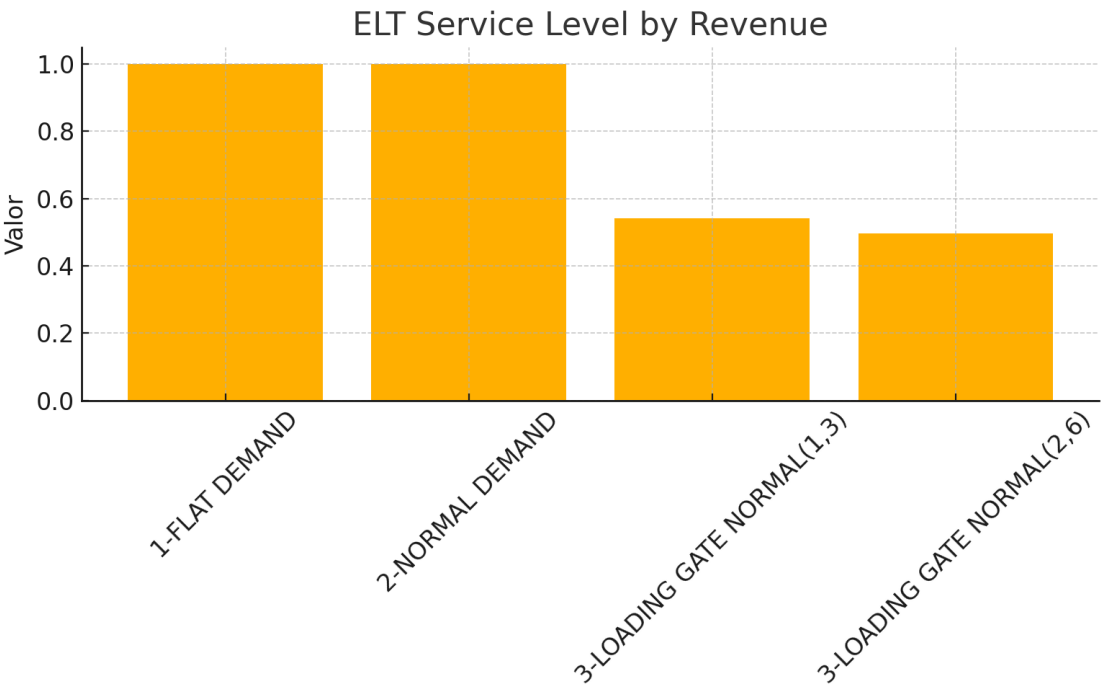
Average Cost per Item



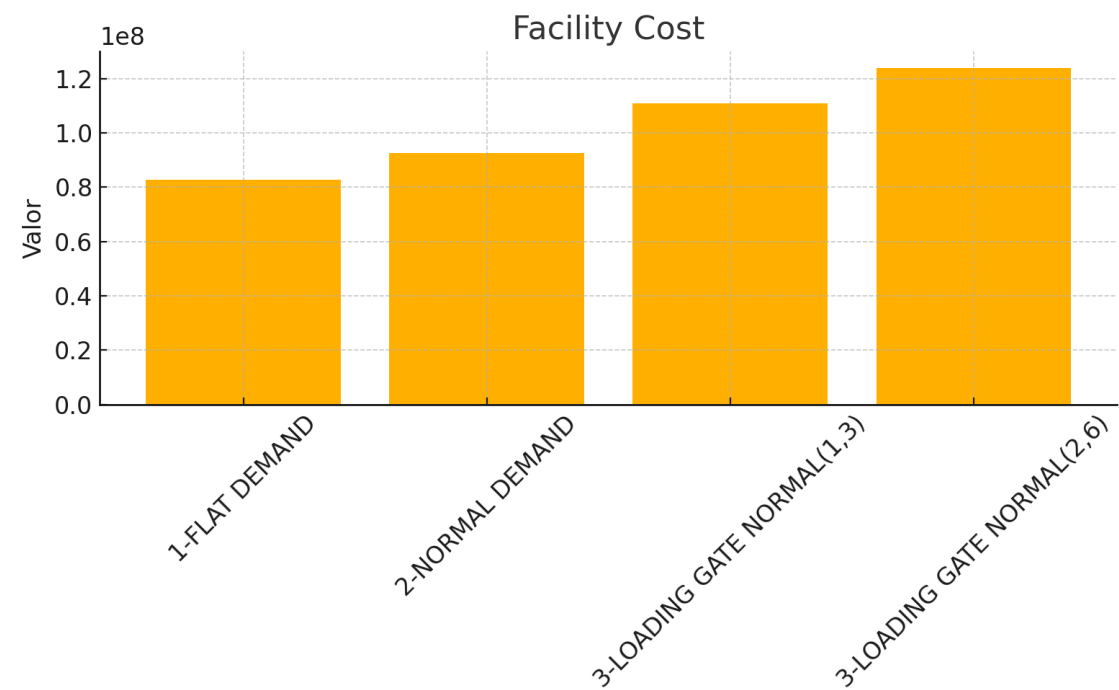
Profit



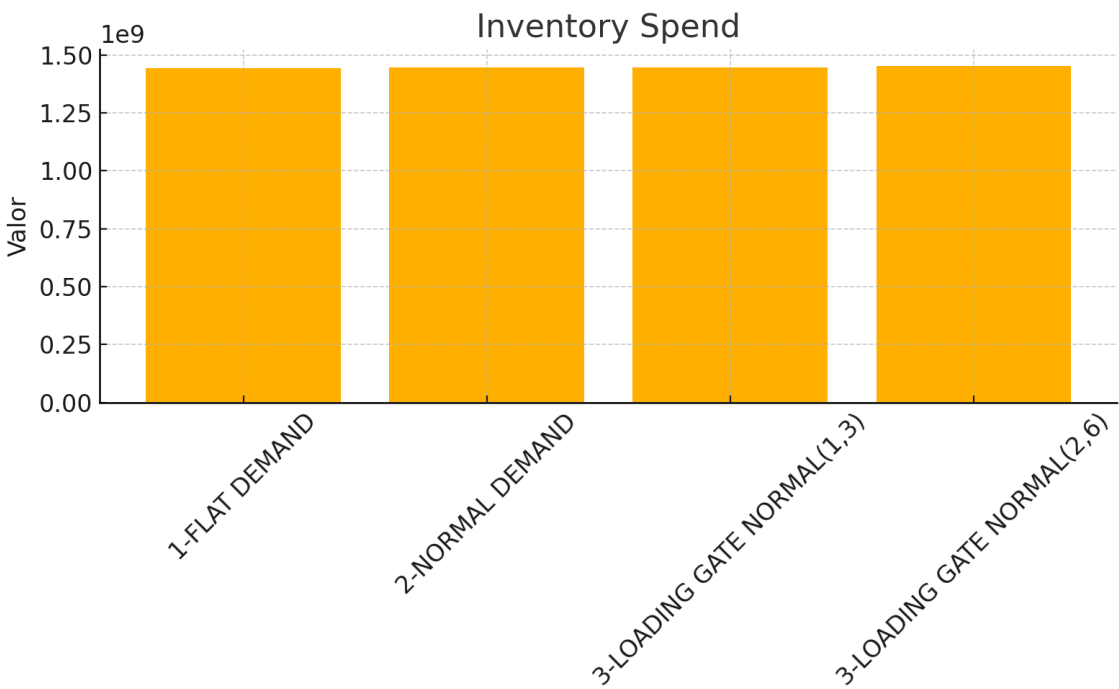
ELT Service Level by Revenue



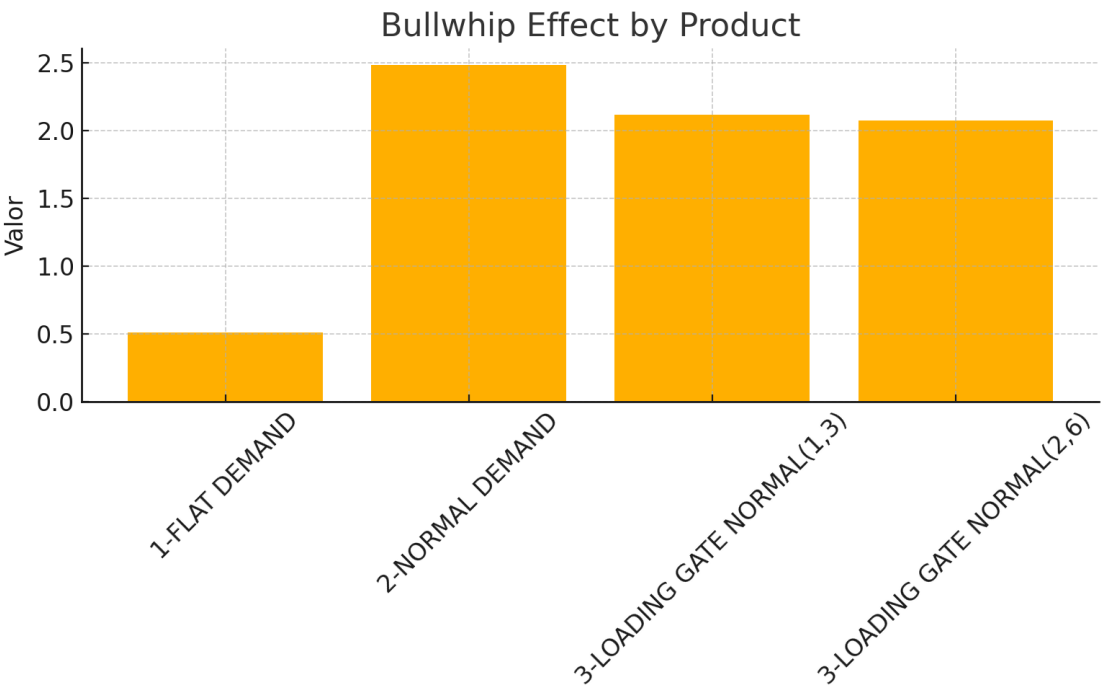
Facility Cost



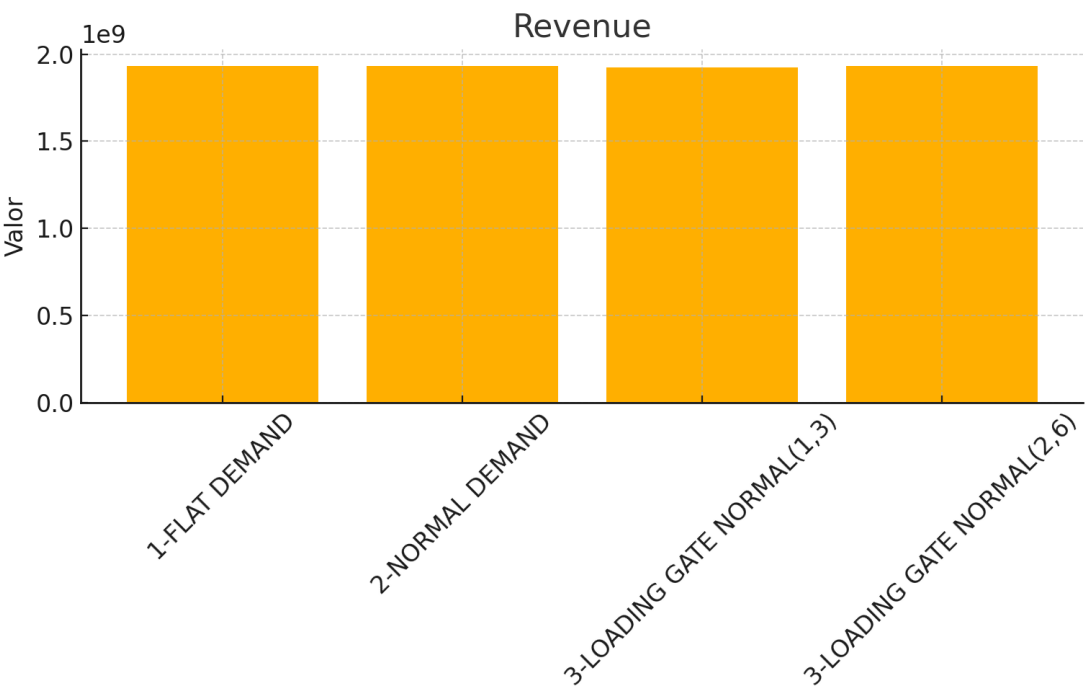
Inventory Spend



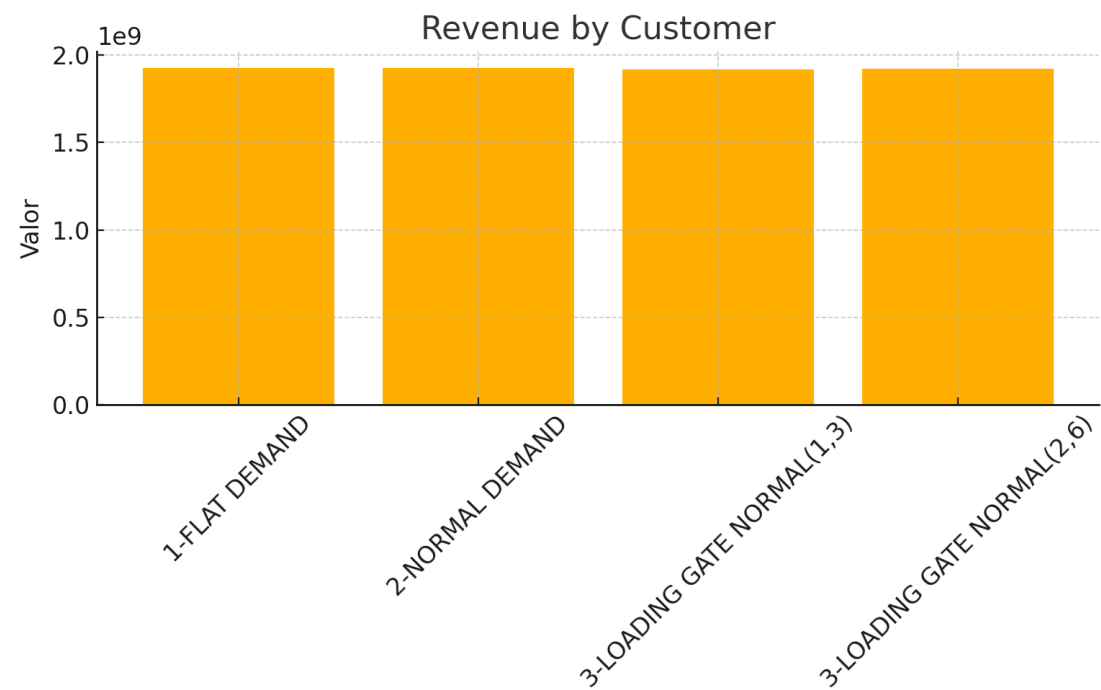
Bullwhip Effect by Product



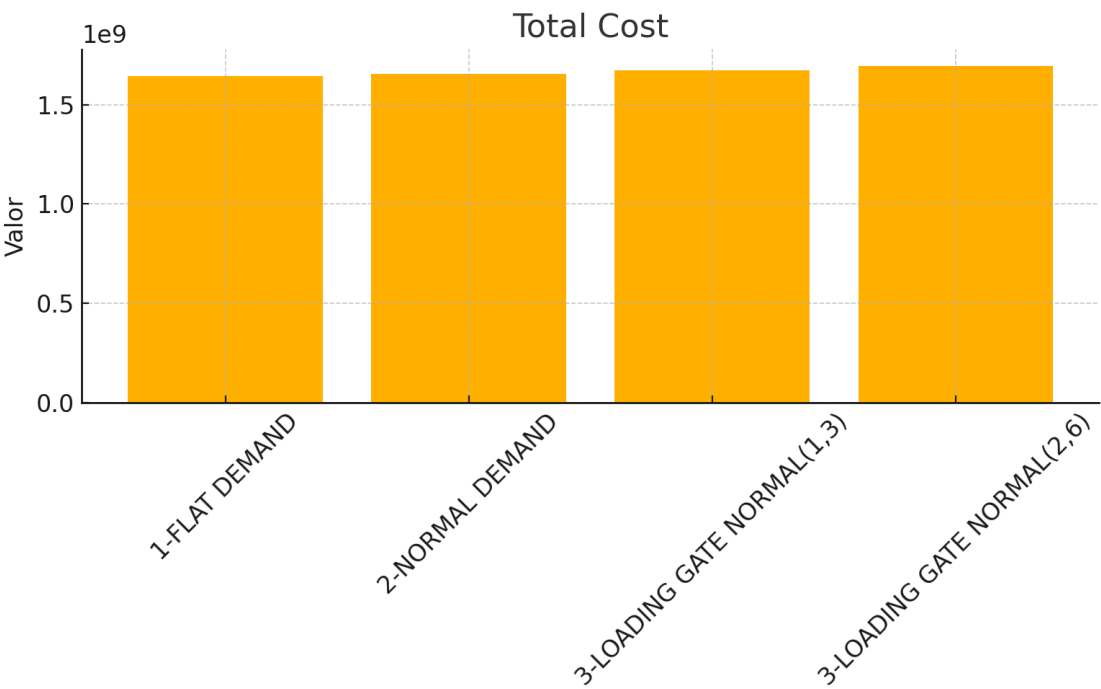
Revenue



Revenue by Customer



Total Cost



Transportation Cost

